requirements.txt - FLIPSIDE

Developer: Gift Zowa
Student Number: ST10443108

Unity Version

Unity 2022.3 LTS

Unity Packages (install via Window > Package Manager)

Package	ID	Purpose
Universal Render Pipeline (URP)	com.unity.render-pipelines.universal	Rendering pipeline
2D Tilemap Editor	com.unity.2d.tilemap	Level building with tilemaps
2D Sprite	com.unity.2d.sprite	Sprite import and slicing
2D Animation	com.unity.2d.animation	Character sprite sheet animation
Input System (New)	com.unity.inputsystem	Keyboard input handling
Cinemachine	com.unity.cinemachine	Smooth camera follow
TextMeshPro	com.unity.textmeshpro	UI text rendering including vertex animation

---

Input Actions Asset

- **File:** FlipsideInputActions.inputactions
- **Action Map:** Player
- **Actions:** Move (Vector2 / WASD + Arrows), Jump (Button / Space), Switch (Button / Tab)

AI-Generated Assets

Code (Cursor - cursor.sh)

All C# scripts were generated and/or debugged with Cursor AI assistance.
Scripts were reviewed, tested, and manually adjusted by the developer before use.

Script	AI Role	Manual Changes

PlayerController.cs	Generated by Cursor	Jump feel tuned, boost outline system added, input forwarding extended
EchoController.cs	Generated by Cursor	No changes needed
CharacterSwitcher.cs	Generated by Cursor	Asymmetric brightness speed added, GetARC/GetECHO methods added
WorldSwitch.cs	Generated by Cursor	Reviewed, no changes
SwitchableObject.cs	Generated by Cursor	Reviewed, no changes
SplitScreenSetup.cs	Generated by Cursor	Reviewed, no changes
CameraFollow.cs	Generated by Cursor	Predictive look-ahead added, double-smoothing Y bug fixed
HazardTile.cs	Generated by Cursor	Flash effect added manually
MovingHazard.cs	Generated by Cursor	Ghost trail and gizmo gradient added
BoostOrb.cs	Generated by Cursor	Coroutine moved to PlayerController to fix duration bug
ExitDoor.cs	Generated by Cursor	Scale punch and continuous pulse added
LevelManager.cs	Generated by Cursor	FreezeAllCharacters added, timer integration added
LevelTimer.cs	Generated by Cursor	Frost icons, vertex wave, strobe effect added
UIManager.cs	Generated by Cursor	Camera shake, label punch, triple-flicker dividing line added
LevelCompleteUI.cs	Generated by Cursor	Bounce easing and elastic header pop added
MainMenu.cs	Generated by Cursor	Title float, subtitle/button animations added

---

Art (Leonardo AI - leonardo.ai)

All visual assets generated using Leonardo Diffusion XL / AlbedoBase XL.
Settings: Guidance Scale 7-8, Alchemy ON, 512x512 output.
Imported into Unity at PPU matching image dimensions, Filter Mode: Point, Compression: None.
Generation prompts fully logged in refinements-changes.md.

Asset	Description
ARC tile set	Ground, wall, corner, hazard spike, background - white/pale blue palette
ECHO tile set	Ground, wall, corner, hazard spike, background - black/neon purple palette
ARC character sprite	Idle, run (x2), jump - white silhouette
ECHO character sprite	Idle, run (x2), jump - black silhouette, neon purple outline
Switch object sprites	Unactivated + activated states, one per world
Boost Orb sprites	Speed (green), Jump (blue), Both (gold) variants
Exit door sprites	Glowing portal, one per world
Moving hazard sprites	Per world variant
Background parallax - ARC	Far + near layer, white/grey geometric
Background parallax - ECHO	Far + near layer, black/purple geometric
Mirror crack dividing line	Full-width UI strip
HUD icons	ARC icon, ECHO icon, Frost icon

Audio (Soundraw / ElevenLabs / GizAI)

Asset	Tool	Description
ARC world music	Soundraw	Calm, minimal electronic ambient
ECHO world music	Soundraw	Tense, low, dark electronic ambient
Tab switch sound	ElevenLabs / GizAI	Sharp click or short whoosh
WorldSwitch activation	ElevenLabs / GizAI	Lever click + world-change whoosh
Boost Orb collect	ElevenLabs / GizAI	Short upward chime
Death / reset sound	ElevenLabs / GizAI	Short negative tone
Level complete chime	ElevenLabs / GizAI	Short positive resolution tone
UI button click	ElevenLabs / GizAI	Soft UI click

Fonts

- **Orbitron** - SIL Open Font Licence (OFL). Free for personal and commercial use.
  Source: Google Fonts - https://fonts.google.com/specimen/Orbitron
  Designer: Matt McInerney

Notes

- No paid Unity Asset Store packages were used
- No third-party copyrighted assets were used without a compatible licence
- All AI outputs were reviewed and deliberately selected before integration
- Asset generation prompts are fully logged in refinements-changes.md

---

References

Anthropic (2024) *Claude* [Large language model]. Available at: https://claude.ai (Accessed: March 2026).

Cursor (2024) *Cursor AI-assisted IDE* [Software]. Available at: https://cursor.sh (Accessed: March 2026).

ElevenLabs (2024) *ElevenLabs AI audio generation* [Software]. Available at: https://elevenlabs.io (Accessed: March 2026).

Google Fonts (2024) *Orbitron* [Typeface]. Available at: https://fonts.google.com/specimen/Orbitron (Accessed: March 2026).

Leonardo AI (2024) *Leonardo.Ai* [Image generation platform]. Available at: https://leonardo.ai (Accessed: March 2026).

Soundraw (2024) *Soundraw AI music generator* [Software]. Available at: https://soundraw.io (Accessed: March 2026).

Unity Technologies (2022) *Unity 2022.3 LTS* [Game engine]. Available at: https://unity.com (Accessed: March 2026).